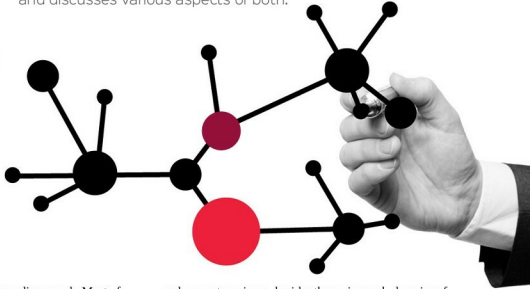


Data Science and Machine Learning: Working Together to Make Machines Smarter

This article clarifies the role of data science in relation to machine learning, and discusses various aspects of both.



Google is now a synonym for online search. Most of us will agree with this, because whenever we try to get information on something we don't know, we say, "Google it!" Have you ever given a thought to how Google comes up with the answers to different questions in an instant? Google and other such search engines make use of different data science algorithms and machine learning techniques to come up with the best results for all our search queries, and that too, in less than a second. Google processes more than 20 petabytes of structured and unstructured data daily and, even then, is able to instantly flash our search results. Had there been no data science and machine learning, Google would not have been able to perform all this and it would not have been the all-pervasive search engine we all depend on today. Data science is one of the roots that support the tree of the digital world.

Data science is also known as data-driven science as it deals with various scientific processes, methods and systems to extract knowledge or insights from large sets of data — either unstructured or structured. We all have access to huge amounts of data, which is about many aspects of our lives – it could be related to communication, shopping, reading the news, searching for information, expressing our opinions, etc. All this is being used to extract useful insights by employing different data science techniques. Data science is basically a concept that unifies statistics with data analysis in order to analyse and relate real world activities with data. It employs different techniques and theories that are drawn from many fields — from within the broad areas of statistics, mathematics, information science

and computer science, besides the various sub-domains of machine learning, cluster analysis, classification, databases, data mining and visualisation.

According to the Turing Award winner Jim Gray, data science is the fourth paradigm of science. Gray asserts that everything about science is rapidly changing because of the impact of information technology and the data deluge. Data science plays a crucial role in transforming the information collected during datafication and adds a value to it. Datafication is nothing but the process of taking different aspects of life and turning these to data. For instance, Twitter datafies different stray thoughts, LinkedIn datafies the professional networks, and so on. We take the help of different data science techniques to extract useful parts out of the collected information during datafication.

Drew Conway is famous for his Venn diagram definition of data science. He applied it to study and analyse one of the biggest problems of the globe – terrorism. If we take a look at his Venn diagram definition, data science is the union of hacking skills, statistical and mathematical knowledge, and substantive expertise about the specific subject. According to him, data science is the civil engineering of data. It requires a practical knowledge of different tools and materials, coupled with a theoretical understanding of what's possible.

The workflow for data science

Data science comprises several sequences of processes which are followed to deduce useful insights from the raw set of